

Department of Science and Humanities

Course Name:	Basics Electrical Engineering(BEE) Course Code: 316U06C104	Semester:	I
Exam:	BEE IA1(Simulation based Assignment)	Division:	All

Q1. Calculate the current through $R1\ \Omega$ resistor using Mesh analysis. In the circuit shown in figure 1, the value of $R1$ will be the **last two digits of roll no. times Ω**

For e.g.: For Roll no: 1000020, then $R1 = 20\Omega$. If your last digits are from 01 to 10, kindly add 10 to your last two digits, so roll no with last two digits with 01 will become 11, then $R1 = 11\Omega$ **(ODD ROLL NO ATTEMPT Q1)**

- Simulate the circuit shown in figure 1, using LT spice software and measure I_{R1} , $I_{4\Omega}$ and $I_{7\Omega}$ and $I_{3\Omega}$.
- Also solve the numerical theoretically and calculate I_{R1} , $I_{4\Omega}$ and $I_{7\Omega}$ and $I_{3\Omega}$.

Tabulate the results as shown in the table below.

Parameter	Theoretical value	Simulated value
I_{R1}		
$I_{4\Omega}$		
$I_{7\Omega}$		
$I_{3\Omega}$		

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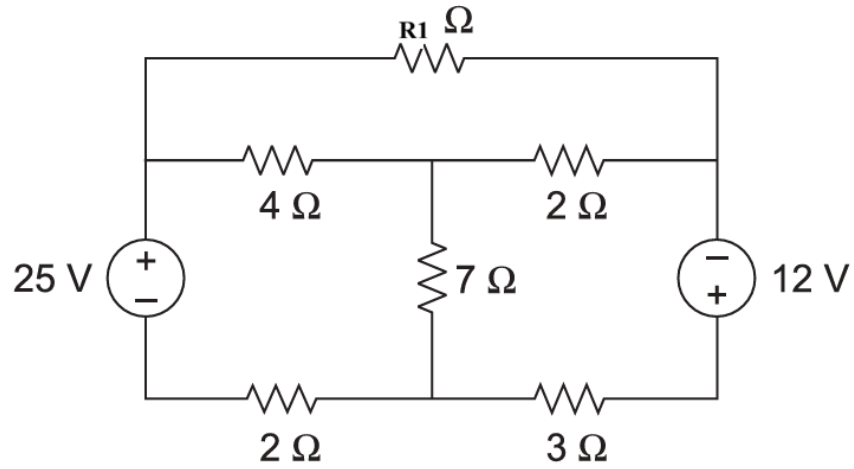


Figure 1

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Q2. Calculate the current through $5\ \Omega$ resistor using Thevenin's theorem. In the circuit shown in figure 2, the value of R_1 will be the **last two digits of roll no. time's Ω**

For eg: For Roll no: 1000020, then $R_1 = 20\ \Omega$. If your last digits are from 01 to 10, kindly add 10 to your last two digits, so roll no with last two digits with 02 will become 12, then $R_1 = 12\ \Omega$ **(EVEN ROLL NO ATTEMPT Q2)**

- Simulate the circuit shown in figure 2, using LT spice software and measure V_{th} , R_{th} and I_L .
- Also Solve the numerical theoretical and measure V_{th} , R_{th} and current through $5\ \Omega$ load resistor (I_L)

Tabulate the results as shown in the table below.

Parameter	Theoretical value	Simulated value
Thevenin's Voltage (V_{th})		
Thevenin's resistance R_{Th}		
Load current I_L		

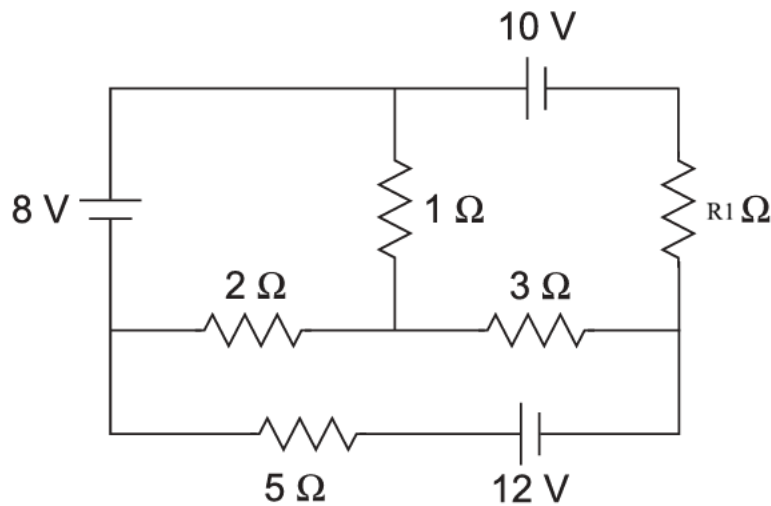


Figure 2